

An aerial photograph of the University of British Columbia Okanagan (UBCO) campus. The image shows several large, modern brick buildings with flat roofs, some featuring solar panels. The campus is surrounded by lush green trees and lawns. In the background, there are rolling hills and mountains under a clear blue sky with a few wispy clouds. The lighting suggests it's either early morning or late afternoon, with long shadows cast across the landscape.

Special Topics: Applied Computer Vision and Deep Learning

UBCO Master of Data Science – DATA 589



Instructure

Mohsen Zardadi, PhD

Lead AI Scientist at TERRASENSE

Research area:

- Applied Computer Vision and AI
- Vision Language Models
- Remote Sensing

Contact: mohsen.zardadi@terrasense.ca always include DATA589 in the subject

Teaching Asistnat

- Ladan Tazik
- Lab Date: Mar 26, Apr 02, Apr 09, Apr 16, Apr 23
- Time: 1:30 to 3:30 p.m.

Grading

iClickers: 10%

Labs: 40%

Quizzes: 50%

Question

Question: How familiar are you with Deep Learning?

- A. I have never used it before
- B. I know a few basic concepts but have not built models myself
- C. I have trained simple models such as image classifiers
- D. I have worked with CNNs or Transformers in projects or coursework
- E. I am very comfortable with Deep Learning and have built or fine-tuned advanced models

Lab Assignments

Weekly lab assignments are worth 40% of your overall grade.

Lab assignments may take more than the two hours lab time.

- No late labs will be accepted.
- A lab may be handed in any time before the due date and may be marked immediately by the TA in the lab.

Lab assignments are done either individually or group (maximum 3 students).

The lab assignments are critical to learning the material and are designed both to prepare you for the exams and build up your skills!

The Quizzes

- There will be **2 quizzes** during the course
- Each quiz is worth **25%** of the final grade
- Together, the quizzes make up **50%** of the total final grade
- Quizzes will assess understanding of key lecture concepts and course

AI Assistants Policy

- Using **AI assistants (e.g., ChatGPT, Copilot, Claude)** is allowed in this course.
- Our policy is the same as working with a **human assistant**: you may ask questions, request explanations, or seek guidance.
- Students are encouraged to **experiment with AI tools**, since many of them are built using deep learning and are relevant to this course.
- However, **AI tools must not be used to complete assignments or homework on your behalf.**
- If you are unsure whether something is allowed, **treat the AI like a human helper and follow the same academic integrity rules.**

Question

Question: Which of the following AI tools have you used during your Master of Data Science program?

- A. ChatGPT
- B. Gemini
- C. Claude
- D. GitHub Copilot
- E. Other

How to Excel in This Course

- Attend lectures and actively participate — many concepts build on each other.
- The best way to learn deep learning is by coding.
- Read notes before class as preparation and try the questions.
- Attend and complete all labs
- Focus on understanding concepts, not just running the code.
- There is tons of online resources if you have time
- Experiment and be curious — try different models, datasets, and ideas.

Course Objectives

- Understand the **fundamentals of computer vision**
- Learn **deep learning architectures for vision** (CNNs, Transformers)
- Build models for **classification, detection, and segmentation**
- Understand the **end-to-end vision pipeline** from data to deployment
- Explore **modern vision systems** such as Vision Language Model, CLIP and generative models

Course References

Textbook:

- Torralba, A., Isola, P., & Freeman, W. T. *"Foundations of Computer Vision"*, MIT Press, 2024

Online Courses & Resources

- Advanced Computer Vision – Mubarak Shah University of Central Florida (UCF)
- Introduction to Computer Vision – Mubarak Shah (UCF)
- Stanford CS231n: CNNs for Visual Recognition
- MIT 6.S191: Intro to Deep Learning
- PyTorch Documentation — pytorch.org

Introduction to Computer Vision (CV)

Introduction to Computer Vision

- Ability of computers to see
 - Image understanding
 - Machine vision
 - Image analysis
 - Video understanding

Image

- 2-D array of numbers (intensity values, gray levels)
- Gray levels 0 (black) to 255 (white)
- Color image is usually 3 2-D arrays of numbers
 - Red
 - Green
 - Blue
- Resolution (number of rows and columns)
 - 128X128
 - 256X256
 - 512X512
 - 640X480
 - 1080X1920



What Machine Sees is just Numbers!

```
img_gray = cv2.imread(image_path, 0)

# Get dimensions from the array shape
height, width = img_gray.shape
# Print the Image Resolution
print("-" * 30)
print(f"IMAGE RESOLUTION: {width} x {height}")
print(f"Total Pixels: {img_gray.size}")
print("-" * 30)

# Display the 2D array (first 5 rows and 5 columns)
print("Image Array Shape:", img_gray.shape)
print("Top-left 5x5 pixel values:\n", img_gray[:15, :15])

# Show the image window
cv2.imshow('Grayscale Image', img_gray)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

```
=====
IMAGE RESOLUTION: 1280 x 720
Total Pixels: 921600
=====

Top-left 15x15 pixel values (The 'Matrix' view):
[[ 0  0  0  0  0  0  0  0  1  1  1  1  1  1  1]
 [117 117 117 117 117 117 117 117 118 118 118 118 118 118 118]
 [109 109 109 109 109 109 109 109 110 110 110 110 110 110 110]
 [114 114 114 114 114 114 114 114 115 115 115 115 115 115 115]
 [115 115 115 115 115 115 115 115 116 116 116 116 116 116 116]
 [112 112 112 112 112 112 112 112 113 113 113 113 113 113 113]
 [115 115 115 115 115 115 115 115 116 116 116 116 116 116 116]
 [117 117 117 117 117 117 117 117 118 118 118 118 118 118 118]
 [117 117 117 117 117 117 117 117 119 119 119 119 119 119 119]
 [117 117 117 117 117 117 117 117 119 119 119 119 119 119 119]
 [117 117 117 117 117 117 117 117 118 118 118 118 118 118 118]
 [117 117 117 117 117 117 117 117 117 117 117 117 117 117 117]
 [117 117 117 117 117 117 117 117 115 115 115 115 115 115 115]
 [117 117 117 117 117 117 117 117 114 114 114 114 114 114 114]
 [117 117 117 117 117 117 117 117 113 113 113 113 113 113 113]]
```



RGB Image

```
=====
IMAGE RESOLUTION: 1280 x 720
CHANNELS: 3 (B, G, R)
=====
```

```
--- BLUE CHANNEL (Top-left 5x5) ---
```

```
[[ 0 0 0 0 0]
 [23 23 23 23 23]
 [28 28 28 28 28]
 [23 23 23 23 23]
 [31 31 31 31 31]]
```

```
--- GREEN CHANNEL (Top-left 5x5) ---
```

```
[[ 5 5 5 5 5]
 [32 32 32 32 32]
 [37 37 37 37 37]
 [32 32 32 32 32]
 [40 40 40 40 40]]
```

```
--- RED CHANNEL (Top-left 5x5) ---
```

```
[[ 2 2 2 2 2]
 [29 29 29 29 29]
 [34 34 34 34 34]
 [29 29 29 29 29]
 [37 37 37 37 37]]
```

```
[]
```



Image Formats

1. Raster Formats (Pixel-Based)	2. Vector Formats (Math-Based)	3. Specialist Formats
• JPEG: best for natural photos; • PNG: supports Alpha (Transparency) channel. • TIFF: industry standard for medical and archival imaging. • WebP/AVIF: dvanced compression formats designed for web delivery. superior compression for high-speed loading.	SVG: Defined by geometric primitives (points, lines, curves). Key Advantage: Infinite scalability without pixelation. Use Case: Icons, logos, and UI elements.	RAW: sensor data before full processing; maximum bit-depth for post-processing.

- Video is sequence of frames
- Usually 24 or 30 frames per second but not always
- Formats: Video formats are slightly more complex than image formats because they are actually made of two parts: a **Container** (the file extension like .mp4) and a **Codecs** (the algorithm used to encode/compress the video).

Containers - Wrapper



The container holds the video stream, audio stream, and metadata (like subtitles) together.



- **MP4 (MPEG-4 Part 14)**: High compatibility. It works on almost every device, browser, and social media platform. It is the best choice for sharing and web use.
- **MOV (QuickTime)**: Developed by Apple. It is a high-quality format preferred by professional video editors because it preserves more detail, though the files are much larger than MP4s.
- **MKV (Matroska)**: An open-source "super-container." It can hold an unlimited number of audio tracks and subtitle languages in one file. It is the favorite for high-quality movie archives.
- **WebM**: Developed by Google specifically for the web. It is royalty-free and designed to provide high quality while using very little data. It is commonly used for YouTube and website background videos.

Codecs (The "Compression Engine")



The codec is what actually shrinks the video so it's not terabytes in size. In your Python code, you might use these to decode a video stream.



- **H.264 (AVC):** The standard and most widely codec for the last decade. It offers a perfect balance between quality and speed.
- **H.265 (HEVC):** The successor to H.264. It is 30%-50% better compression than H.264, meaning you get the same quality at half the file size. This is what makes **4K and 8K** streaming possible.
- **AV1:** The newest major player. It is open-source and provides up to **50% better compression** than H.264. Giant platforms like Netflix and YouTube are moving toward this to save bandwidth.

Question

Question: Which statement best explains the difference between a video container and a codec?



- A. A container compresses the video, while a codec stores subtitles
- B. A codec is the file extension, while a container is the image resolution
- C. A container holds video/audio/metadata together, while a codec encodes or compresses the video stream
- D. A codec only works for images, while a container only works for audio
- E. A container and codec mean exactly the same thing



Hugging Face is a platform and open-source ecosystem for sharing, discovering, and using machine learning models and datasets.



What Hugging Face Provides

- Model Hub – Thousands of pretrained models (vision, NLP, audio, multimodal)
- Datasets Hub – Public datasets ready for research and experimentation
- Transformers Library – Easy-to-use tools for loading and running models
- Community & Documentation – Tutorials, demos, and open research projects

Common Computer Vision Tasks



Image-Level Tasks

- **Image Classification** – Assign a label to an entire image
- **Image Retrieval** – Find similar images in a database



Object-Level Tasks

- **Object Detection** – Locate and classify objects using bounding boxes
- **Instance Segmentation** – Identify and segment each object instance

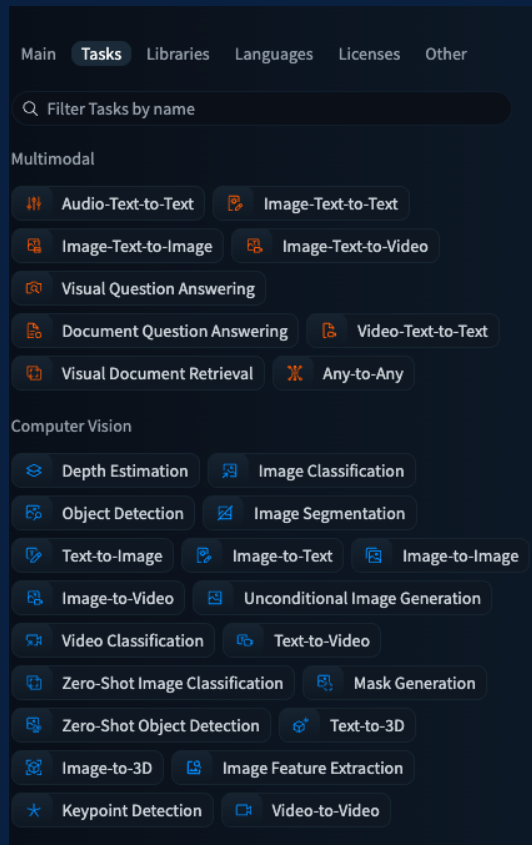
Pixel-Level Tasks

- **Semantic Segmentation** – Assign a class label to every pixel
- **Depth Estimation** – Predict distance from the camera for each pixel

Video-Level Tasks

- **Object Tracking** – Track objects across video frames
- **Action Recognition** – Recognize activities or events in videos

CV Tasks on Hugging Face



Object Recognition



Automatically identifying and understanding objects present in images or videos.

Subcategories of Object Recognition



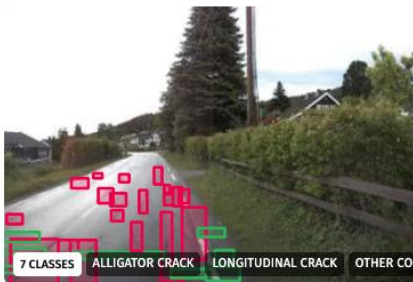
- **Image Classification:** What object(s) are in the image? one or more class labels
- **Object Detection:** What objects and where are they? class labels + bounding boxes
- **Instance Segmentation:** Distinguish each individual object. pixel-level masks per object
- **Semantic Segmentation:** Assign a class to every pixel. Does not separate individual instances

Object Detection



Object Detection
UAVDT

78K images · 836K labels
Added 2024-03-05 · 31K views



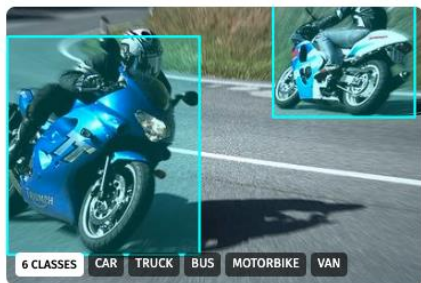
Object Detection
RDD2022

47K images · 61K labels
Added 2024-01-22 · 29K views



Object Detection
KITTI Object Detection

15K images · 52K labels
Added 2023-11-27 · 25K views



Object Detection
Vehicle Dataset for YOLO



Object Detection
VisDrone2019-DET



Object Detection
Drone Dataset (UAV)

Object Classification



A snapshot of two root-to-leaf branches of ImageNet: the top row is from the mammal subtree; the bottom row is from the vehicle subtree. ImageNet: A Large-Scale Hierarchical Image Database , Deng + CVPR 2009

Semantic Segmentation



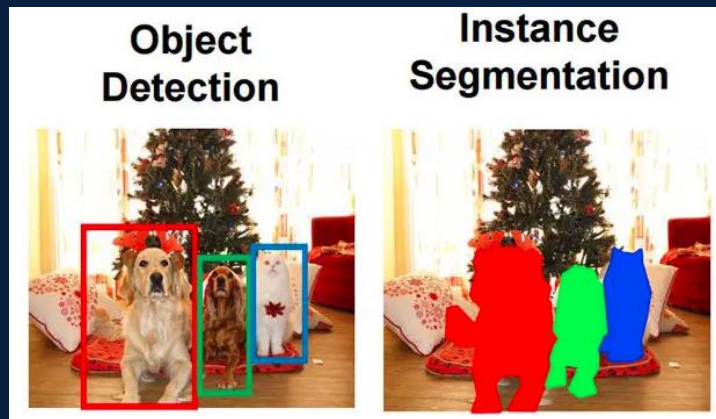
- Assigns a class label to every pixel in an image.
- Pixels belonging to the same category share the same label.
- Does not distinguish between different instances of the same object.



Microsoft Common Objects in Context (COCO) Dataset

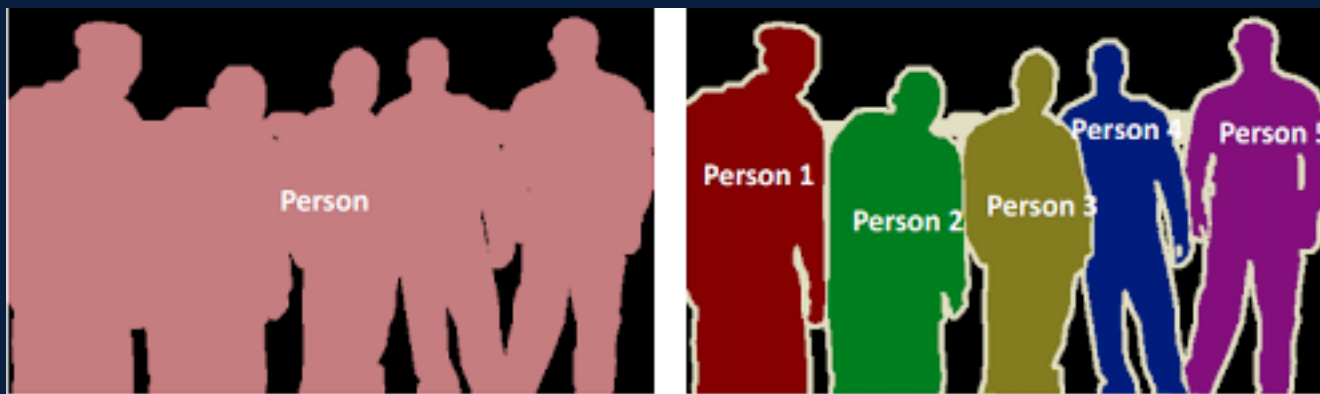
Instance Segmentation

- Assigns a **class label** to every pixel and separates **different object instances**.
- Each object of the same category receives a **different mask**.



Medium: Car image segmentation using Convolutional Neural Nets
By [Ashish Malhotra](#)

Semantic vs Instance Segmentation



Folio3.ai

Applications of Computer Vision

- **Autonomous Navigation & Systems:** The "frontier" of CV, encompassing Self-Driving Cars, UAVs (Drones), and Deep-Sea RPAS, requiring real-time SLAM and path planning.
- **Video Surveillance & Tracking:** Robust multi-object tracking (MOT), crowd analysis, and human behavior recognition.
- **Remote Sensing:** Analysis of satellite and aerial imagery for urban planning, agriculture, and climate monitoring.
- **Medical & Biological Imaging:** Precision diagnostics via tumor segmentation, automated cell pathology, and AI-guided robotic surgery.
- **Spatial Intelligence (AR/VR):** The bridge between graphics and vision, using SLAM and 3D reconstruction to merge digital data with physical geometry.
- **Human-Centric Vision:** Beyond simple face recognition; includes pose estimation, gait analysis for security, and emotional/behavioral monitoring.
- **Industrial & Visual Analytics:** Automated quality control (inspection), satellite-based remote sensing, and intelligent video surveillance for anomaly detection.



Computer Vision and Pattern Recognition – CVPR(2026)

- 3D from multi-view and sensors
- 3D from single images
- Active learning
- Adversarial attack and defense
- Affinity groups
- **Autonomous driving**
- Biometrics
- Computational imaging
- Computer vision theory
- Deep learning architectures and techniques
- Document analysis and understanding
- Efficient, edge, and scalable vision
- Egocentric vision
- **Embodied vision: Active agents, simulation**
- Event-based cameras
- Explainable computer vision
- **Foundation models (LLM, VLM, VLA, etc.)**
- Generative models
- Human modeling & understanding: Face, body, pose, gesture, movement
- **Image and video synthesis and generation**
- Low-level vision
- Machine learning (other than deep learning)
- Medical and biological vision, cell microscopy
- **Multimodal detection, recognition, segmentation**
- Multimodal learning
- Reinforcement learning and reasoning
- Open world learning
- Optimization methods (other than deep learning)
- Photogrammetry and remote sensing
- Physics-based vision and shape-from-X
- Recognition: categorization, detection, retrieval
- Representation learning
- Robot perception
- Scene analysis and understanding
- Segmentation, grouping and shape analysis
- Self-, semi-, meta- and unsupervised learning
- **Synthetic data for vision**
- Transfer/low-shot/ continual/ long-tail learning
- Transparency, safety, fairness, accountability, and ethics in vision
- Video: Action and event understanding
- Video: Low-level analysis, motion, and tracking
- Vision and graphics
- Vision for accessibility
- Vision for healthcare
- Vision for privacy and security
- Vision for scientific discovery
- Vision for societal good
- Vision for XR, AR, VR
- Vision, language, and reasoning
- Vision applications and systems
- World models
- Emerging topics - other

Core & Emerging Directions in CV

Computer Vision is no longer a single task (classification/detection); it is a system combining perception, reasoning, simulation, learning efficiency, and decision-making.

- Embodied Vision & Robotics

Vision for action: perception, manipulation, navigation, and human–robot interaction

- Autonomous Systems & Self-Driving Vision

Real-time, safety-critical perception with multi-sensor fusion and temporal reasoning

- Digital Twins & Synthetic Data

Simulation-driven learning, scalable data generation, and sim-to-real transfer

- Multimodal & Foundation Vision Models

Vision-language models, promptable perception, open-vocabulary and zero-shot learning

- Data-Centric & Efficient Learning

Improving model performance through better data, active learning, and continual learning

- Vision-Based AI Agents

Systems that see, reason, plan, and act in interactive environments

- Trustworthy & Robust Vision Systems

Reliability, uncertainty estimation, robustness, and deployment-ready CV

Applied CV Projects

Remote Sensing Equipment:



- Side Looking Airborne Radars - SLAR
- Infrared / Ultraviolet Line Scanners
- Electro-Optical Infrared Camera Systems – EO/IR
- Automatic Identification Systems
- Satellite Communications Systems
- Geo-coded Digital Camera Systems

Applied CV Projects at TerraSense

National Aerial Surveillance Program

NASP

- Preventing pollution
- Whale Detection
- Search and Rescue



Applied CV Projects

MX15

<https://youtu.be/MOUuxlTR4eA?si=-8qasBUyw5JQu968>

Synthetic Aperture Radar (SAR)



Synthetic Aperture Radar - SAR

SAR imagery can see through smoke, fog and at night.



<https://airadar.com/drone-deployed-sar/>

Question

Question: Which computer vision task assigns a class label to every pixel but does not separate different objects of the same class?



- A. Image classification
- B. Object detection
- C. Instance segmentation
- D. Semantic segmentation
- E. Object tracking

Datasets



When we looked at where authors chose to host their datasets, a clear pattern emerged – over 80% of the accepted papers used a handful of widely adopted platforms: **Hugging Face, Kaggle, Dataverse, and OpenML**. Another 13% relied on self-hosted or bespoke solutions, with the rest distributed across smaller repositories like Zenodo and the Open Science Framework.



NeurIPS 2025- <https://blog.neurips.cc/2025/12/05/neurips-datasets-benchmarks-track-from-art-to-science-in-ai-evaluations/>

Dataset Platforms

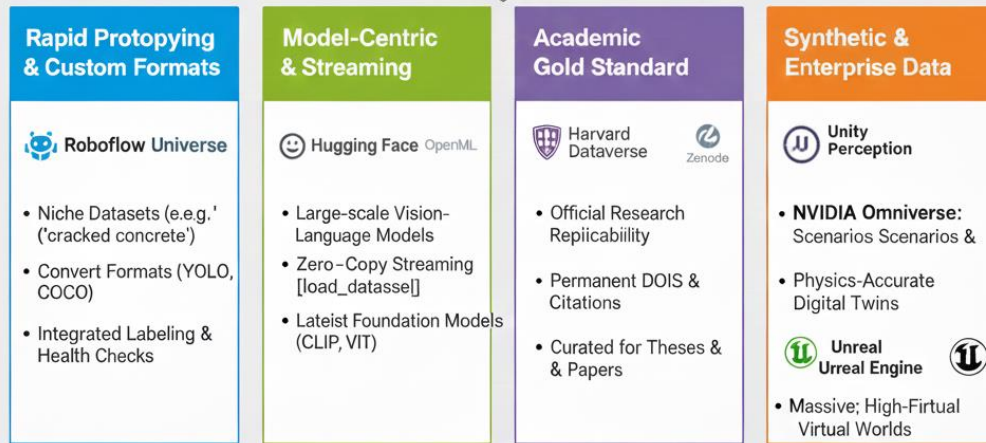
The world has moved away from "Unzip and Pray."

- **2015:** You downloaded a .zip from a professor's website, and half the labels were missing.
- **2026:** You use a **Platform** that provides a "Data Health Check." It tells you before you download: "10% of your images are blurry" or "Class A has 1000 images but Class B only has 10."

Platform choice = Workflow speed.

Where do I find data for my CV project?

Computer Vision Dataset Platforms: Decision Guide (2026)



CV Dataset Links

- <https://datasetninja.com/>
- <https://universe.roboflow.com/>
- <https://www.earthdata.nasa.gov/>
- <https://datasetsearch.research.google.com/>
- <https://datahub.io/>
- <https://www.kaggle.com/datasets>

Synthetic/Simulated data



What is synthetic data?

Synthetic data is data (images/videos) (and labels) generated by simulation or generative models instead of manual collection.



Key idea: you can generate perfect ground truth (boxes, masks, depth, optical flow, pose, 3D keypoints) at scale.

Why?

Labeling is the bottleneck: real-world data is expensive, slow, and often incomplete (and privacy-sensitive).

Synthetic gives you:

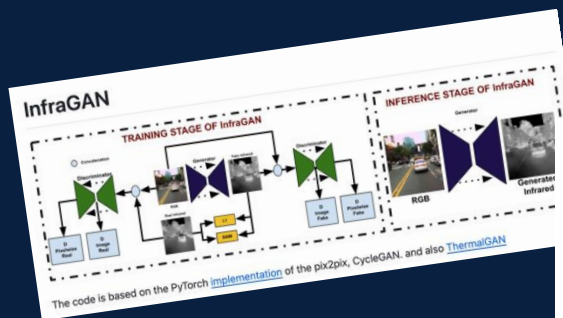
- Scale (millions of labeled frames)
- Coverage (rare events: accidents, edge cases, unusual lighting/weather)
- Control (balanced classes; targeted scenarios)
- Safety & privacy (human-centric data without collecting real identities)
- Synthetic is especially strong when you need long-tail or edge cases.

Renderings vs Generative

Physically Based Rendering (3DS Max, Omniverse, Unreal Engine, Blender)

Generative Methods (CycleGAN, pix2pix, diffusion)

- Data-driven (needs lots of data)
- Easier to scale



ThermalGAN

This is the PyTorch implementation of the color-to-thermal image translation presented on ECCV 2018 in the paper.

The code is based on the PyTorch [implementation](#) of the pix2pix and CycleGAN.

ThermalGAN: [\[Project\]](#) [\[Paper\]](#)



I2V-GAN

This repository is the official PyTorch implementation for ACM MM 2021 paper "I2V-GAN: Unpaired Infrared-to-Visible Video Translation" [\[Arxiv\]](#) [\[ACM DL\]](#)

Traffic I2V Example:

Download a pretrained model from [Baidu Netdisk](#) [Access code: Trxf] or [Google drive](#)



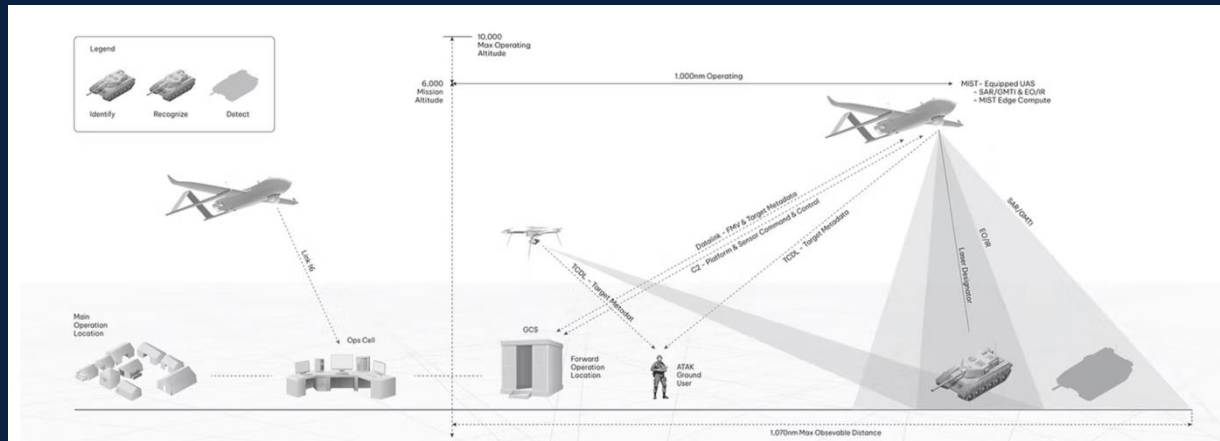
Real EO -> Synthetic IR – Car Chase



Synthetic Data Tools

- **Python** (data generation scripts, dataset conversion, QA)
- **CV formats:** COCO, YOLO, VOC; depth/pose formats; multi-sensor sync
- **3D basics:** coordinate frames, camera intrinsics/extrinsics, projection
- **Simulation / synthetic platforms (choose one stack first)**
- **Omniverse / Isaac Sim / Replicator** (OpenUSD pipeline; robotics + synthetic). [NVIDIA Developer+1](#)
- **Unity + Perception** (randomization framework + labelers; CV training workflows). [arXiv+2Unity Documentation+2](#)
- **AirSim (Unreal)** (autonomy simulation + data generation). [GitHub+2Microsoft+2](#)
- **Nice-to-have (advanced)**
- **Blender** (procedural generation, rendering pipelines)
- **Diffusion models** for augmentation / style alignment

TerraSense: Airborne Edge Processing and Target Detection



MIST transforms airborne **Intelligence, Surveillance, and Reconnaissance (ISR)** by processing sensor data in real time at the edge, reducing bandwidth demands while increasing certainty. By integrating feeds from EO/IR, WAMI, radar, and other onboard sensors, it automatically detects, tracks, and classifies ground targets while filtering noise, motion clutter, and false positives. Operators receive only the clearest and most actionable intelligence, enabling faster decisions and safer missions.

TerraSense - MIST



CV Conferences



- *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*
- *International Conference on Computer Vision (ICCV)*
- *European Conference on Computer Vision (ECCV) – from 1990*
- *International Conference on Pattern Recognition (ICPR) - Lyon, France August, 17-22, 2026*
- *Asian Conference on Computer Vision (ACCV) - December 14-18 in osaka, japan*
- *IEEE International Conference on Image Processing – ICIP*



Objectives

- Understand **what computer vision is** and its role in modern AI systems
- Explain how **images are represented as numerical arrays** (pixels, RGB, resolution)
- Distinguish between **image formats, video containers, and codecs**
- Identify major **computer vision tasks** such as classification, detection, and segmentation
- Understand the difference between **semantic segmentation and instance segmentation**
- Become familiar with **Hugging Face and modern AI model hubs**
- Recognize common **datasets, benchmarks, and synthetic data approaches** used in computer vision



THE UNIVERSITY OF BRITISH COLUMBIA

